MIL-P-13537(Ord)
13 July 1954
SUPERSEDING
FCP 1516
11 February 1953
AFPS 145
16 August 1944

MILITARY SPECIFICATION

PACKAGING OF BOARDS, PLOTTING, M5 AND M5A2

1. SCOPE

1.1 This specification covers the preservation, packaging, packing and marking of Boards, Plotting, M5 and M5A2 with equipment, for overseas shipment or domestic shipment involving extended storage, domestic shipment for limited storage, and domestic shipment for immediate use.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

NN-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

JAN-P-100	Packaging and Packing for Overseas Shipment - General Specification
JAN-P-106	Boxes; Wood, Nailed (Overseas type)
MIL-P-116	Preservation, Methods of
JAN-B-121	Barrier Materials, Greaseproof
JAN-P-127	Packaging and Packing for Overseas shipment - Tape, Adhesive, Pressure Sensitive, Water-Resistant
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier Material
MIL-A-148	Aluminum Foil
MIL-C-3133	Cellular Rubber Products, General Purpose (Except Ebonite)
MIL-L-3150	Lubricating Oil, Preservative, Medium
MIL-G-3278	Grease; Aircraft and Instrument (For Low and High Temperatures)
MIL-C-5092	Cement, Rubber (Synthetic and Reclaimed Rubber Base)
MIL-L-10547	Liners, Case, Waterproof
MIL-C-11796	Compound, Corrosion Preventive, Petrolatum Type, Hot Application

U. S. Army

3-182	Compound, Insulation, Ignition
19-10	Filler, Interior Packing, Fiberboard, Corrugated, Single and Double Faced

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.

STANDARD
Military
MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-The following shall apply wherever packaging materials are referred to in this specification by name only:

Reference	Full Identification
Greaseproof barrier material	Type I or II, Grade A or C, class 1 or 2 or Specification JAN-B-121
Pressure sensitive tape	Any type and grade of Specification JAN-P-127
Water-resistant Adhesive	Specification MIL-A-140

3.2 Overseas shipment or domestic shipment involving extended storage.- Where plotting boards are to be shipped overseas or for domestic shipment involving extended storage, the following disassembling, cleaning, preserving, packaging, packing and marking shall apply:

3.2.1 Disassembling and cleaning.

3.2.1.1 Disassembling.-The drafting machine assembly (arm) shall be removed from the support. The spindle assembly shall be disconnected and the frame removed from the support. The thumb screws securing the frame to the legs shall be loosened and the support removed. The thumb screws securing the brace to the legs shall be loosened and the legs removed.

3.2.1.2 Cleaning.-The exterior surfaces of the plotting board shall be cleaned so as to be free from dirt, smudges, fingerprints and other foreign substances in accordance with Methods C-1 and C-4 of Specification MIL-P-116. Method C-4 shall be used to clean unprotected metal surfaces. The packing chest shall be cleaned in accordance with Method C-1.

3.2.2 Preserving.

3.2.2.1 The unpainted surfaces of the spindle assembly shall be coated with preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116).

3.2.2.2 The spanner wrench for Board, Plotting M5A2 and the two spanner wrenches for Board, Plotting M5 shall be coated with preservative conforming to class 2 of Specification MIL-C-11796 (P-5 of Specification MIL-P-116).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.

3.2.2.3 Preservative conforming to Specification MIL-G-3278 (P-11 of Specification MIL-P-116) shall be applied on the exposed surfaces of the bearings of the support.

3.2.3 Packaging.--The plotting board in the packing chest shall be packaged Method I as specified herein and illustrated in figures 1, 2 and 3.

3.2.3.1 The coated surfaces of the spindle assembly shall be wrapped with greaseproof barrier material, which shall be secured in place with pressure sensitive tape. The wrapped spindle assembly shall then be positioned in the chest as shown in figure 2.

3.2.3.2 The coated spanner wrenches shall be wrapped with greaseproof barrier material, which shall be secured with pressure sensitive tape. The wrapped wrenches shall then be positioned in the chest.

3.2.3.3 The brace and legs shall be positioned and secured in their respective positions in the chest (see figure 2).

3.2.3.4 The spare screws for Plotting Board M5A2 shall be wrapped with greaseproof barrier material and placed in the receptacle provided in the chest.

3.2.3.5 The coated surfaces of the bearings of the support shall be covered with aluminum foil conforming to Specification MIL-A-148 (at least 0.002 inch thick). The barrier material shall be secured with pressure sensitive tape. The support then shall be positioned in the chest as shown in figure 2.

3.2.3.6 The surface of the frame which contacts the bearings shall be covered with a piece of double face corrugated fiberboard (approximately 26 inches in diameter) conforming to A flute of Specification 19-10. The fiberboard shall be positioned and secured to the frame with pressure sensitive tape as shown in figure 2. The frame shall then be positioned in the chest as shown in figure 3.

3.2.3.7 The drafting machine assembly shall be positioned and secured in the lid of the chest by means of the supports and blocks which are provided in the lid.

3.2.3.8 The instruction book shall be placed in the compartment provided for the book in the chest.

3.2.3.9 The lid then shall be closed and secured with the trunk bolts, hasp and keeper, and padlock. After locking the padlock, the keys shall be removed from the padlock and secured to the chain, which is attached to the padlock, with wire in such a manner that the keys cannot become lost.

3.2.4 Packing.

3.2.4.1 The packing chest containing the plotting board shall be packed in a shipping container containing a waterproof case liner as specified herein and illustrated in figure 4.

3.2.4.2 The shipping container shall conform to either of the following specifications:

PPP-B-601	Style A or B, type II or III, class 2
JAN-P-106	Style 2 or 4.

3.2.4.3 The steel straps shall not be stapled on the shipping container for domestic shipment involving extended storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.

3.2.4.4 The panel of the shipping container, which will be adjacent to the side of the chest containing the lock, hasp and keeper, shall have two cleats (one in each corner) nailed to the inside surface with the width of the cleat contacting the panel and the length of the cleat parallel with the vertical corner. Each cleat shall be 13 inches long by 4 inches wide by ½ inch thick.

3.2.4.5 Each shipping container shall contain a case liner (in which the packing chest is packed) conforming to type I, grade A or B, class 2 of Specification MIL-L-10547.

3.2.4.6 Each corner of the chest shall contain a corner support fabricated from wood and cellular rubber as shown in figure 5. The cellular rubber shall conform to grade RS-14 (preferred) or grade RS-13 (at least ¼ inch thick) of Specification MIL-C-3133, and shall be secured to the wood surfaces with cement conforming to type 2 of Specification MIL-C-5092.

3.2.4.7 Two fiberboard pads shall be placed inside the case liner for added support for the chest. Both pads shall be positioned approximately in the center of the chest with the length of the pads parallel to the width of the shipping container. One pad shall be secured to the bottom of the case liner with water-resistant adhesive before the chest is placed inside the case liner. After the chest is positioned inside the case liner, the other pad shall be placed on top of the chest. Water-resistant adhesive shall be applied on the top of the pad so that the case liner will adhere to the pad when the case liner is closed and sealed. Each pad shall be fabricated from commercial solid fiberboard and shall be approximately 40 inches long x 4 inches wide x 1 inch thick.

3.2.4.8 The following inside dimensions, cubage and gross weight are approximate for one unit packaged plotting board, packed in a nailed wood box:

Inside dimensions	- 47-3/8 inches long x 46-5/8 inches wide x 13-½ inches deep
Cubage	- 21 cubic feet
Gross weight	- 345 pounds

3.2.5 Marking.-

3.2.5.1 The information to be placed on each case liner and shipping container shall be in accordance with Standard MIL-STD-129. (See 6.2 for identification data to be added.)

3.2.5.2 Protective coatings on markings.-All markings on the shipping container shall be coated with compound conforming to Specification 3-182.

3.3 Domestic shipment for limited storage.-Where plotting boards are to be shipped for domestic shipment for limited storage, the following disassembling, cleaning, packaging, packing and marking requirements shall apply:

3.3.1 Disassembling and cleaning.-Disassembling and cleaning shall be as specified in 3.2.1.

3.3.2 Preserving.-Preservative shall be applied on the bearings as specified in 3.2.2.3.

3.3.3 Packaging.-Packaging shall be as specified in 3.2.3 except that greaseproof barrier material, specified in 3.2.3.1 and 3.2.3.2 may be omitted. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.

aluminum foil, specified in 3.2.3.4 shall be used to cover the bearings.

3.3.4 Packing.

3.3.4.1 The packing chest containing the plotting board shall be immobilized in a shipping container conforming to Specification NN-B-621 or PPP-B-601.

3.3.4.2 Wood blocks shall be nailed to the inside surfaces of the side and end panels of the shipping container in such a manner that the handles, trunk bolts, lock, etc. on the chest are at least ½ inch from the wood surfaces of the container.

3.3.5 Marking.-Marking shall be as specified in 3.2.5, except that "PACKED FOR LIMITED STORAGE" shall be plainly stencilled on one end and one side panel of each shipping container. The method of protection to be included in the identification data shall be "Method III."

3.4 Domestic shipment for immediate use.-Where plotting boards are to be shipped for domestic shipment for immediate use, the requirements contained in 3.3 shall apply, except that "PACKED FOR IMMEDIATE USE" shall be plainly stencilled on one end and one side panel of each shipping container and the serial number in the identification data may be omitted.

3.5 Sample pack.-A preproduction pack shall be made in accordance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.-The preproduction pack shall be inspected and approved by a Government inspector prior to packaging and packing the items for shipment. Production packs shall be tested at random where the Government inspector considers it necessary to conduct tests to assure conformance with the requirements of this specification.

4.2 Inspection.-All methods and materials shall be subject to inspection by a Government inspector.

4.3 Test procedures.-At the discretion of the Government inspector, the preproduction pack and any production packs selected for test, shall be tested in accordance with Specification MIL-P-116 and Specification JAN-P-100.

5. PREPARATION FOR DELIVERY

(Not applicable)

6. NOTES.

6.1 Ordering data.-Invitation for bids and contracts or orders should specify the title, number and date of this specification and the type of shipment involved.

6.2 Identification data.-

6.2.1 The following indicates the stock number to be added in the identification data for the plotting board:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.

ITEM	STOCK NO.
Board, Plotting, M5 W/E Composed of: 1 Board, Plotting M5 1 Chest, Packing 2 Wrench, Spanner 1 Instruction Book	F001-6702975
Board, Plotting, M5A2, W/E Composed of: 1 Board, Plotting, M5A2 1 Chest, Packing 1 Wrench, Spanner 2 Spare Screws 1 Instruction Book	F001-6702971

6.2.2 The serial number should be added to the identification data.

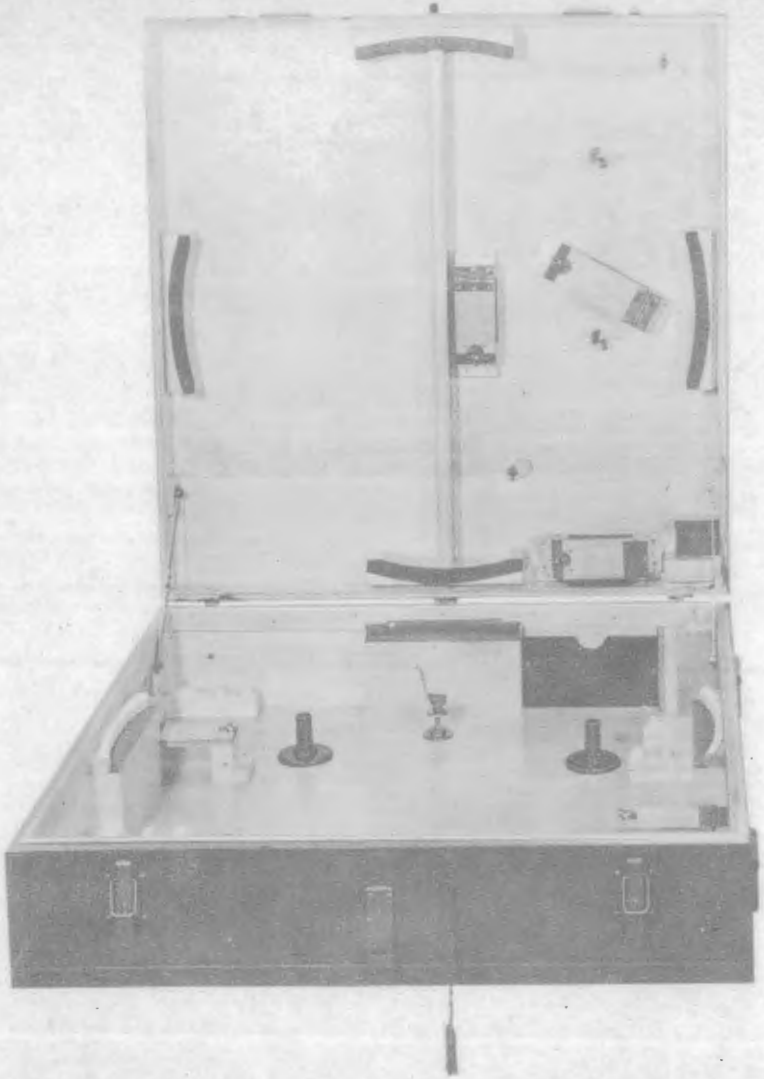
6.2.3 The method of protection also should be included in the identification data.

PATENT NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or sell any patented invention that may in any way be related thereto.

CUSTODIAN:

Army-Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.



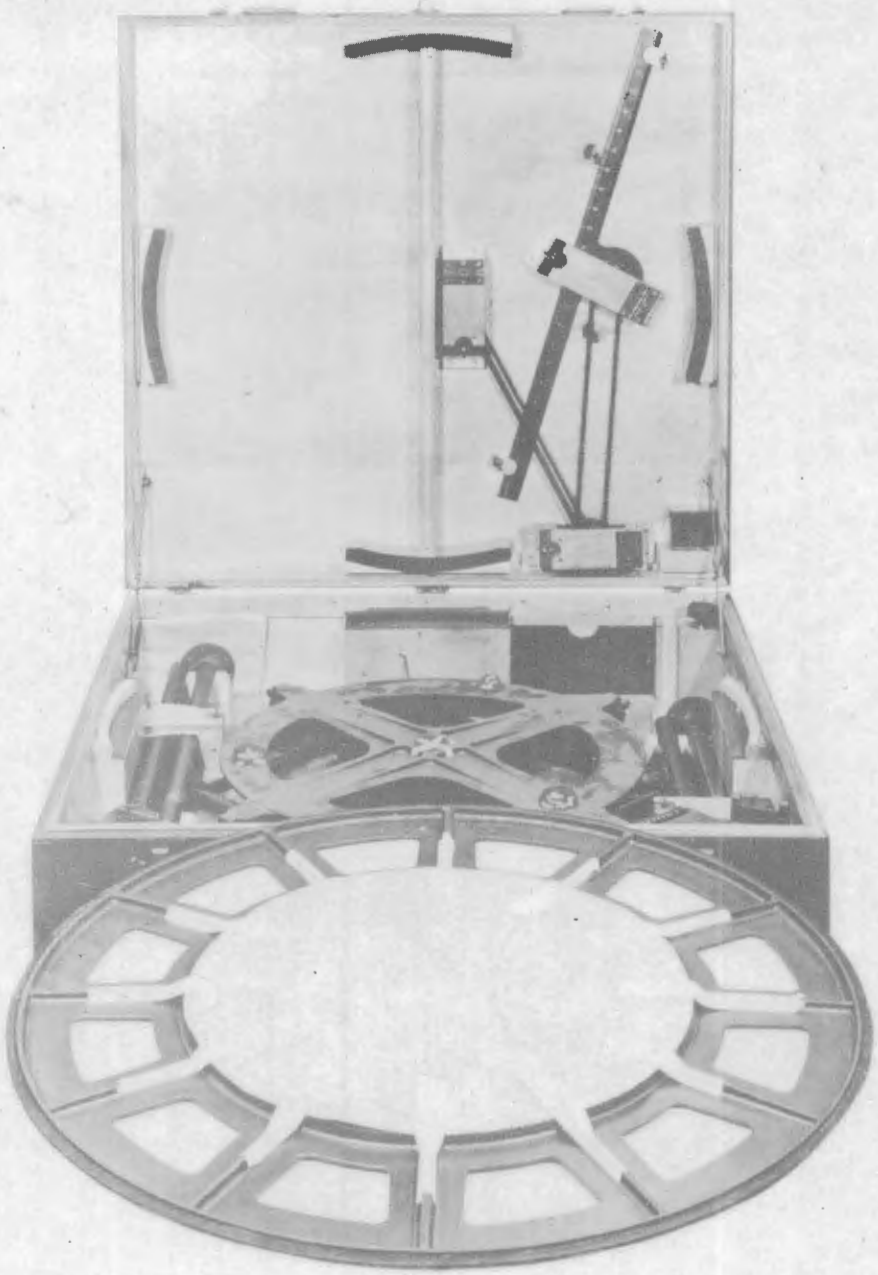


NEG. #23788-1 5-20-53 FRANKFORD ARSENAL ORD. CORPS

PLOTTING BOARD M5A2, W/E - Showing chest, assembled plotting board, spare screws, manual and wrench.

FIGURE 1

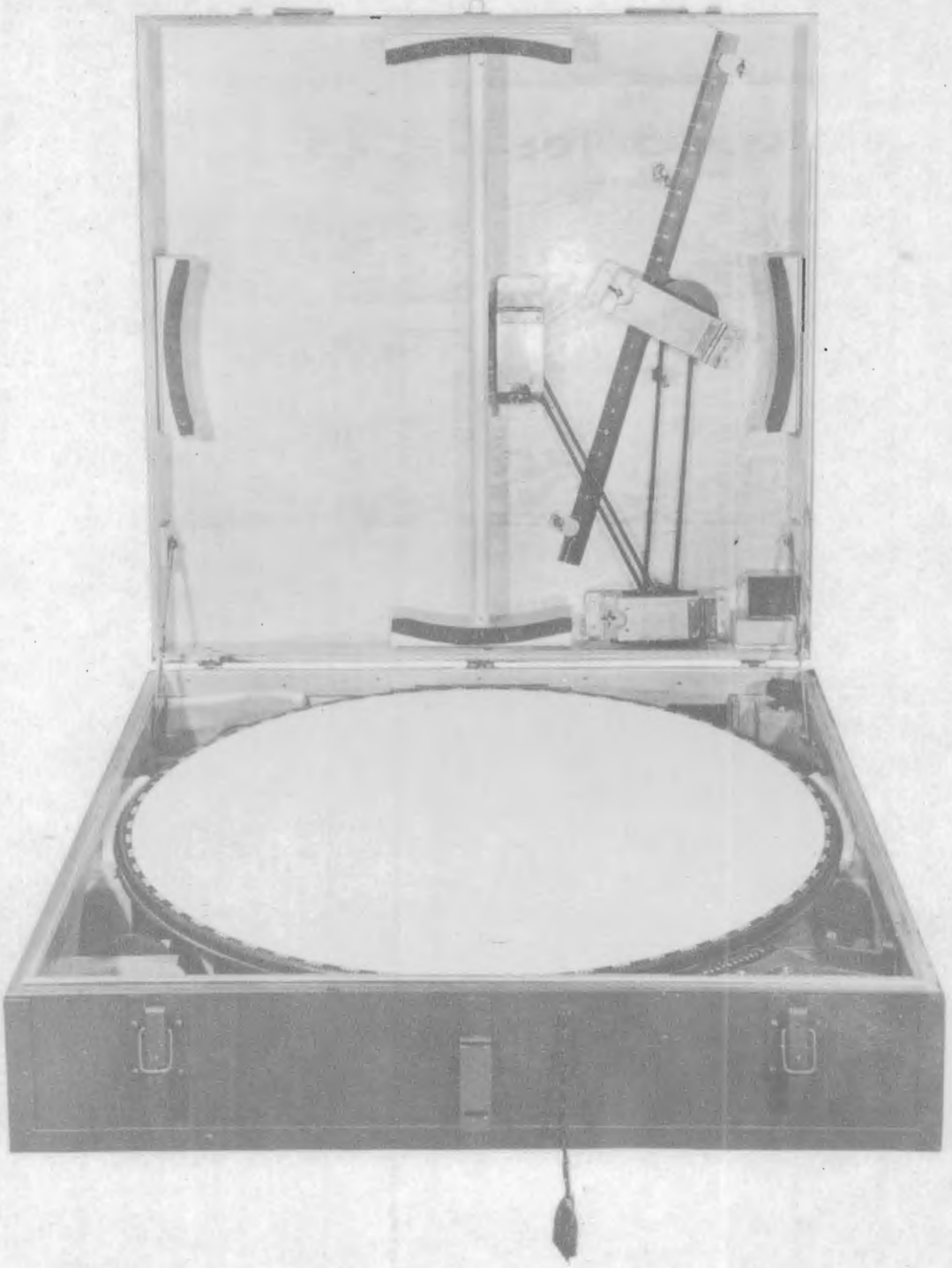
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.



NEG. #23788-2 5-20-53 FRANKFORD ARSENAL ORD. CORPS

PLOTTING BOARD M5A2, W/E - Packing the disassembled plotting board in the chest.
FIGURE 2

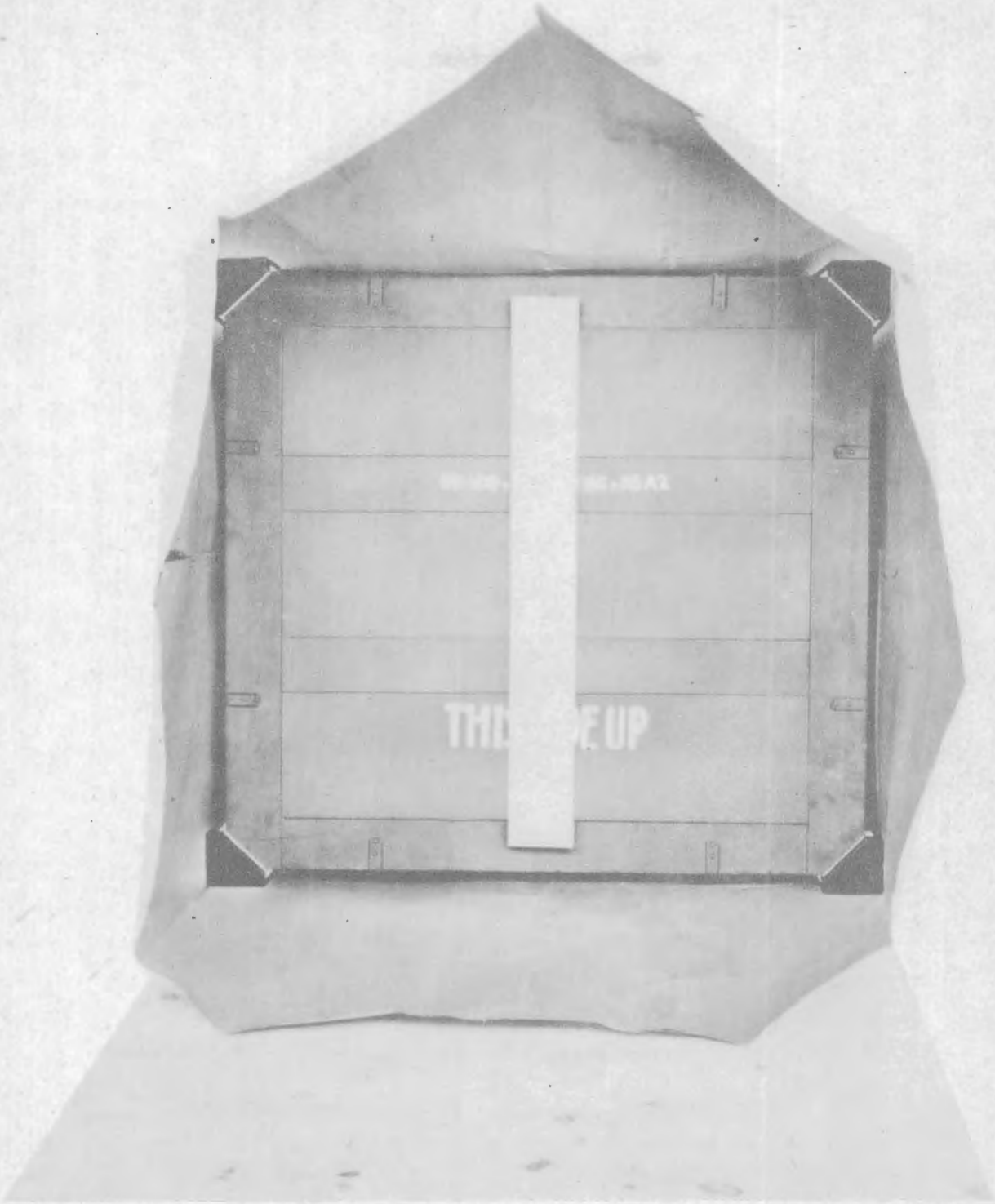
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.



NEG. #23788-3 5-20-53 FRANKFORD ARSENAL ORD. CORPS

PLOTTING BOARD M5A2, W/E - Plotting board in packing chest.
FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.



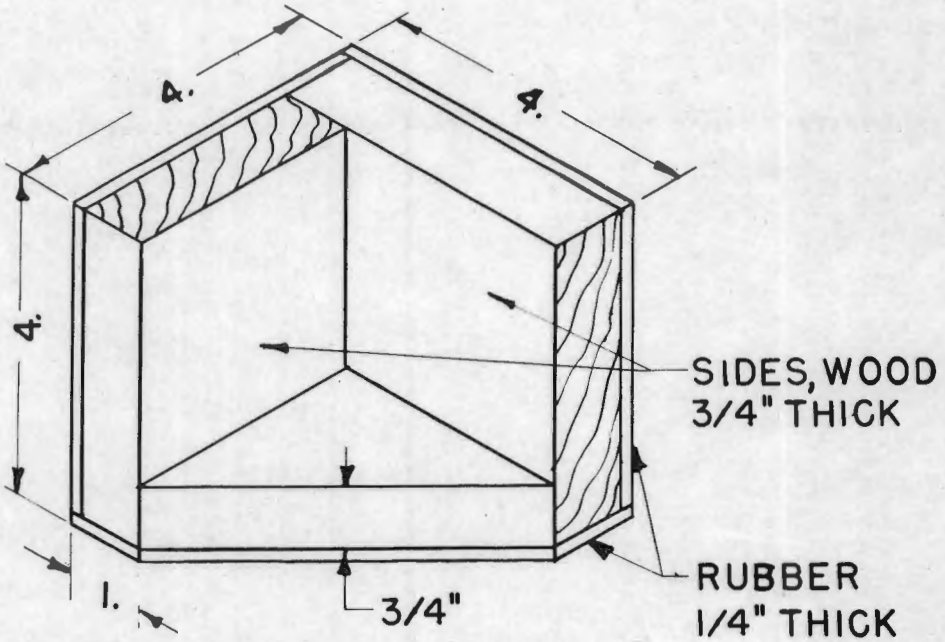
NEG. #23788-4 5-20-53 FRANKFORD ARSENAL ORD. CORPS

PLOTTING BOARD M5A2, W/E - Chest immobilized within a case liner inside the shipping container.

FIGURE 4

MIL-P-13537(Ord)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.



CORNER SUPPORT
8 REQUIRED

FIGURE 5

NEG. #23788-5 5-20-53 FRANKFORD ARSENAL ORD. CORPS

MIL-P-13537

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:49Z
Check the source to verify that this is the current version before use.